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User's Guide for the NASA High Efficiency Centrifugal Compressor Data Archive

Herbert Monroe Harrison III
Glenn Research Center, Cleveland, Ohio

This Revised Copy, numbered as NASA/TM-20240003692/REV1, December 2024, supersedes the previous version, NASA/TM-20240003692, May 2024, in its entirety.

December 2024

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National Aeronautics and
Space Administration

Glenn Research Center
Cleveland, Ohio 44135

December 2024

Acknowledgments

Please contact Trey Harrison by email at herbert.harrison@nasa.gov for inquiries related to the dataset and documentation or to be added to an email list to be notified of updates and additions to the archive. Any revisions or corrections made to the archive will be documented in the revision history.

This effort is sponsored by the Transformational Tools and Technologies project of the NASA Transformative Aeronautics Concepts Program and the Revolutionary Vertical Lift Technology Project of the NASA Advanced Air Vehicles Program. The HECC Data Archive has grown to include contributions from numerous government, industry, and academic organizations, and the author is grateful to all of the individuals who have supported the efforts along the way.

Major contributors to the archive are Tammy Nguyen-Huynh of both the NASA Glenn Research Center and the Ohio State University and Qingyuan Zhuang at Siemens. Tammy's work with the HECC impeller solid model led to the discovery of the geometry differences between the As-Manufactured and Design-Intent geometries. Qingyuan Zhuang conducted the cold-to-hot analysis of the As-Manufactured impeller and generously shared the Hot As-Manufactured impeller coordinates with NASA for hosting on the Data Archive and use by the community.

RTX Technologies Research Center conducted the design work on the HECC from its inception and released the vaneless diffuser set for use by the community. Alex Camargo, Jacob Jaksic, Matthew Blaha, Jonathan Mitchell, Carolyn Thiemann, and Jozsef Puskas have spent innumerable hours acquiring data on the HECC at the Small Engine Components Compressor Test Facility at the NASA Glenn Research Center. David Na helped find the appropriate tools to set up the publicly accessible Data Archive.

Revised Copy

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User's Guide for the Nasa High Efficiency Centrifugal Compressor Data Archive

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SUMMARY

The data sets contained in this archive are associated with the High Efficiency Centrifugal Compressor (HECC) in the Small Engine Components Compressor Test Facility, colloquially referred to as CE-18, at the NASA Glenn Research Center. The archive is accessible at https://storage.googleapis.com/hecc-data/NASA-HECC-Data-Archive.zip. The documentation contained herein provides context for the data hosted on the archive. The data sets and accompanying content in this document will be updated periodically as additional data is acquired and analyzed. The revision date of this document is indicated in the footer, and the revision updates are noted in the Revision History. Contact Trey Harrison (herbert.harrison@nasa.gov) for inquiries related to the data set and documentation or to be added to an email list to be notified of updates and additions to the archive.

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1. REVISION HISTORY

March 2023	Initial publication of archive with HECC vaned diffuser, vaneless diffuser, and transition duct data at design tip clearance.
August 2023	• Included notes about geometry discrepancies between models and blade sections (Section 4.) • Provided HECC impeller cold blade sections (Section 6.1.2) • Added HECC vaneless diffuser solid model (Section 7.6 Models)
February 2024	• Section 4.6 Design-Intent and As-Manufactured Geometry Differences updated • Added tip clearance data to the archive and Read-Me • Section 7.3 Data – Transition Duct Inlet updated to include relevant impeller exit survey metadata • Publications added to HECC Data Archive
July 2024	• Updates throughout the archive to clarify geometry, in particular to Section 4.6 Design-Intent and As-Manufactured Geometry Differences • Geometry nomenclature updated for clarity • Added inlet survey data (Section 6.3 Data – Stage Inlet Surveys) • Added Turbo Expo 2024 publications
October 2024A	• The tip blade sections for the both the impeller main blade and splitter blade were identified as missing from the archive. The missing sections have been added to the *HECC As-Manufactured Impeller Coordinates (Cold, With Fillets)* directory.
October 2024B	• Updates made to the humidity correction procedures for the HECC Vaned Diffuser Data Set (Section 6); information regarding the updates are in Section 4.4 Gas Properties • Geometric parameters compiled for reference in Section 5.1 Geometric Parameters
December 2024	• Document rearranged for clarity; section names updated • Added As-Manufactured Impeller Hot Geometry Coordinates to the *HECC As-Manufactured Impeller Coordinates (Hot, With Fillets)* directory; documentation in Section 6.1.3 HECC As-Manufactured Impeller Coordinates (Hot, With Fillets) • Added Section 4.2 Nomenclature and Instrumentation Overview • Updated Section 4.3 Bleed Flows and Metal Temperatures to document metal temperatures • Aerodynamic design point conditions compiled in Section 5.2 Design Point Aerodynamic Conditions

2. NOMENCLATURE

b	blade height
c_p	specific heat at constant pressure
D	diameter
$\dot{m}$	mass flow rate
N_c	corrected speed
r	radius
T	temperature
U	wheel speed
ρ	density
Φ	inlet flow coefficient
ψ	Loading coefficient

2.1 Acronyms

BLM	baseline metal (inlet)
BLP	baseline plastic (inlet)
CAD	computer aided design
CFD	computational fluid dynamics
EGV	exit guide vane
FLA	forward looking aft
HECC	High Efficiency Centrifugal Compressor
MB	main blade
MH	modified hub (inlet)
TPR	total pressure ratio
TDC	top dead center

3. INTRODUCTION

The High Efficiency Centrifugal Compressor (HECC) was developed with the following goals according to Medic et al. [1]:

- Identify key technical barriers in the advancement of state-of-the-art of small centrifugal compressor stages.
- Delineate measurements required to provide insight into the flow physics of the technical barriers.
- Design, fabricate, install, and test a state-of-the-art research compressor that is representative of the rear stage of an axial-centrifugal aero-engine.
- Acquire detailed aerodynamic data to clarify flow physics and establish extensive data sets for future application.

As of the present revision, two different diffuser configurations and three different inlet configurations are available for the HECC. The originally developed vaned diffuser featured aggressive design targets intended to advance state-of-the-art compressor technology; the follow-up vaneless diffuser test was developed to better understand the impeller performance in isolation from the vaned diffuser. The three inlet configurations are the (1) baseline metal inlet, (2) baseline plastic inlet, and (3) modified-hub plastic inlet. The baseline metal and baseline plastic inlets have exactly the same geometry, but as suggested by their names, are manufactured from metal and plastic, respectively. The modified-hub inlet features the same shroud contour as the baseline inlets, but the hub curvature is more aggressive. The modified-hub inlet was developed to understand the effect of an aggressive S-duct configuration on centrifugal compressor performance.

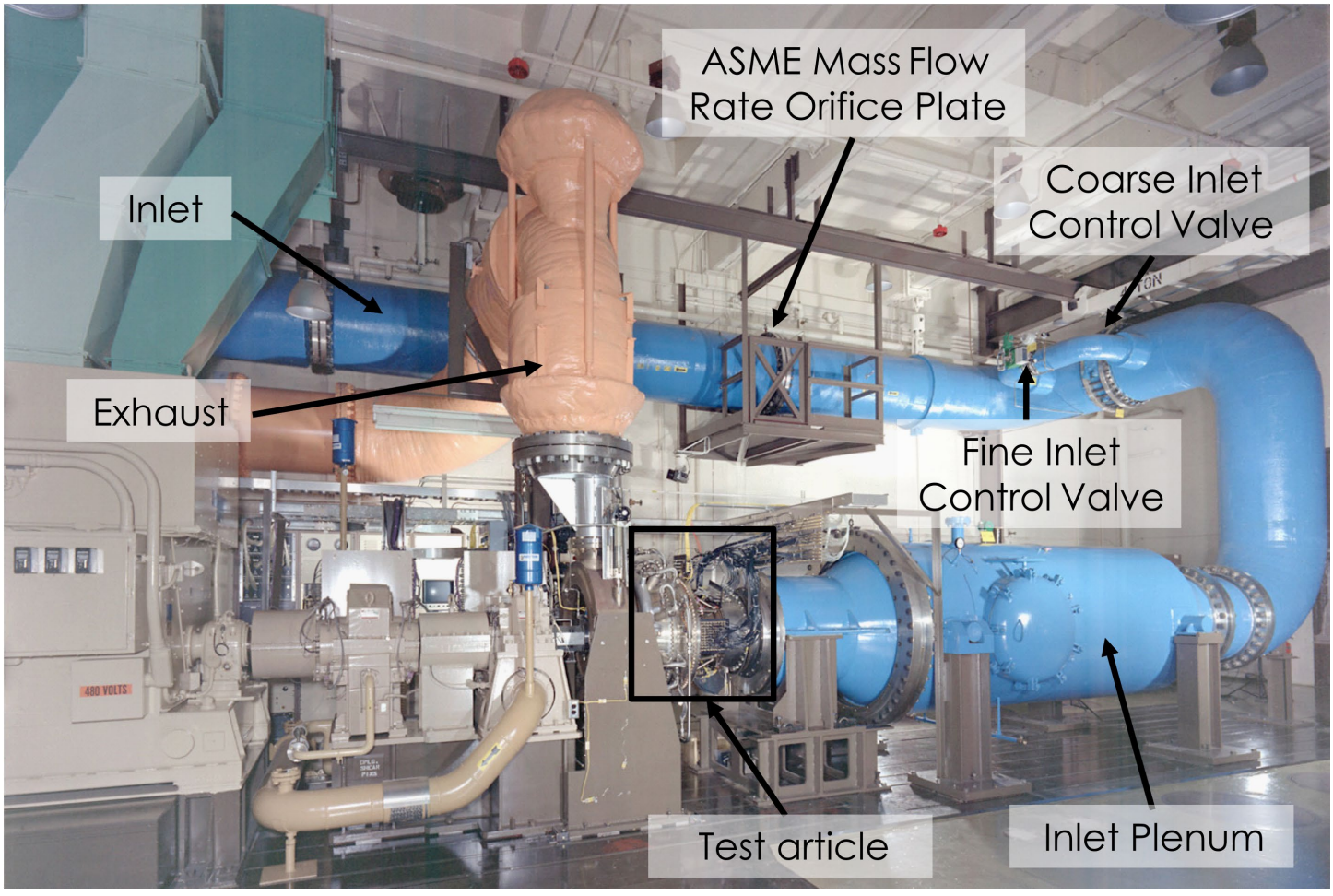


FIGURE 1: MAJOR FLOWPATH COMPONENTS OF THE NASA SMALL ENGINE COMPONENTS COMPRESSOR TEST FACILITY.

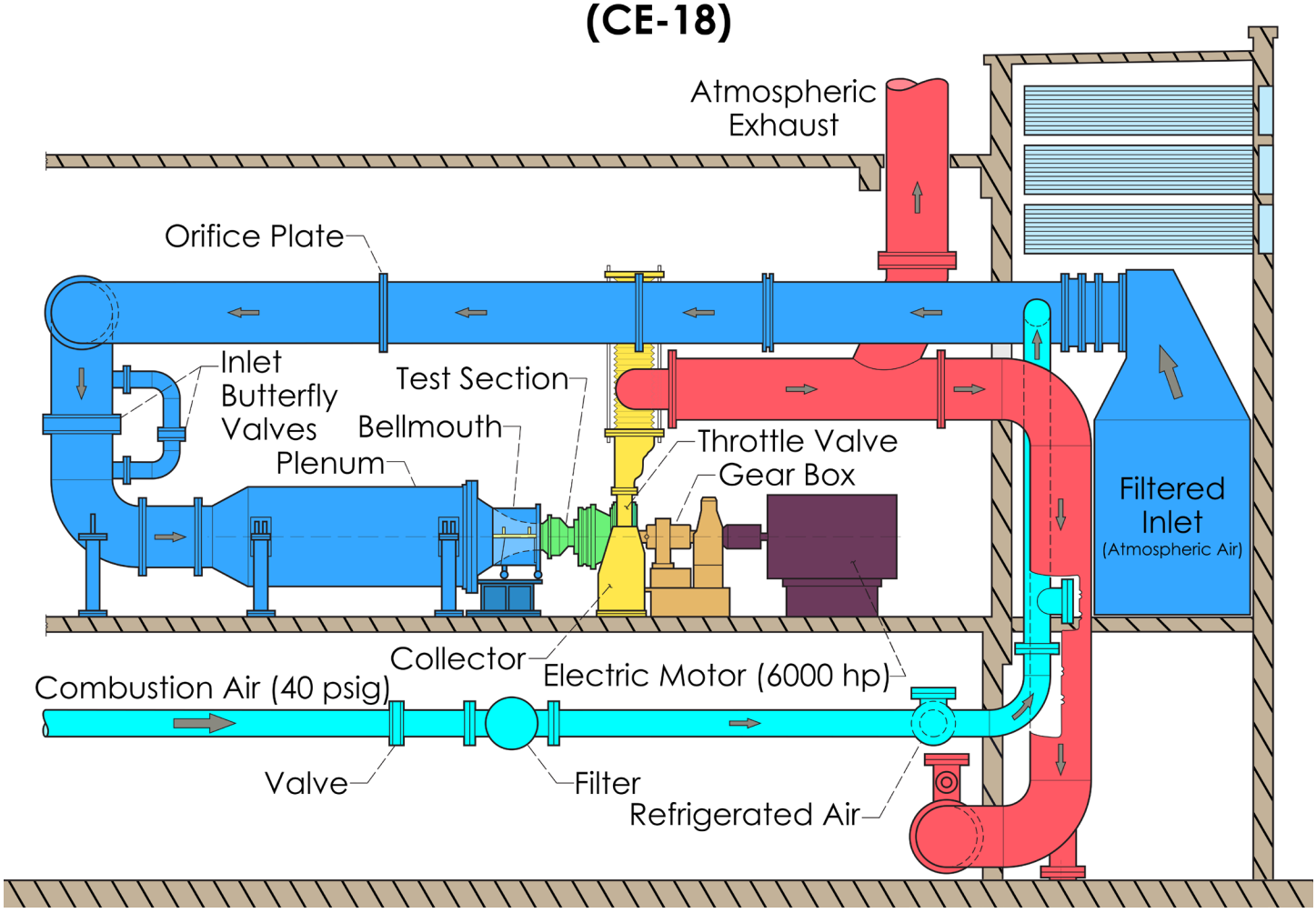


FIGURE 2: MAJOR COMPONENTS OF THE NASA SMALL ENGINE COMPONENTS COMPRESSOR TEST FACILITY (CE-18).

The HECC test article has been housed in the NASA Small Engine Components Compressor Test Facility since the test article's inception. The details of the facility are documented in detail by Brokopp and Gronski [2] and Harrison et al. [3]. A photograph of the facility and accompanying schematic are provided for reference in Figure 1 and Figure 2, respectively. Figure 3 shows the HECC with the extensive instrumentation and auxiliary systems connected and in test-ready condition.

Extensive documentation of the HECC vaned diffuser stage design and performance can be found in the report published by Medic et al. [1]. Braunscheidel et al. [4] give a briefer review of the vaned diffuser configuration with less emphasis on the design process but include additional aerodynamic data. A facility report published in 2023 by Harrison et al. documents the present state of the Small Engine Components Compressor Test Facility and vaneless diffuser performance with all three inlet configurations [3]. Robles Vega et al. [5], Robles Vega et al. [6], and Harrison et al. [7] review the Design-Intent and As-Manufactured impeller geometries in detail and extensively investigate the effect of the size of the tip gap on the impeller performance and flow physics. All publications relevant to the HECC published with NASA affiliation are included in the *Publications* directory.

In the interest of brevity, the details of each of those reports are not contained herein. Instead, this document serves as a tool for navigation and utilization of the open experimental data associated with the HECC test article. The following sections outline the conventions,

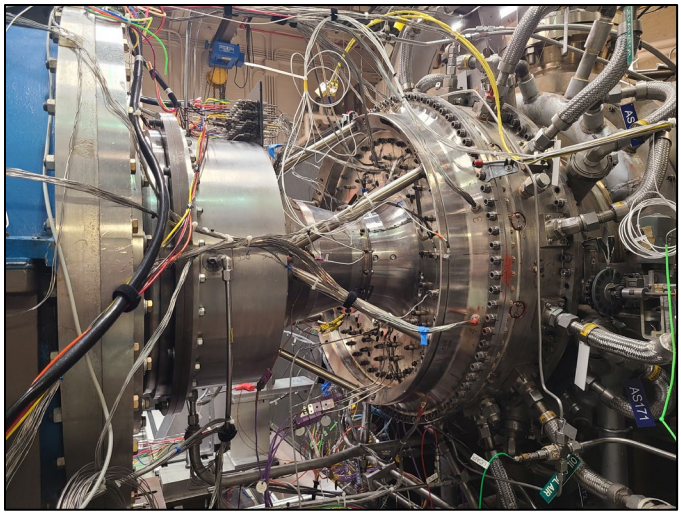


FIGURE 3: HECC TEST ARTICLE WITH ALL INSTRUMENTATION AND AUXILIARY SYSTEMS CONNECTED.

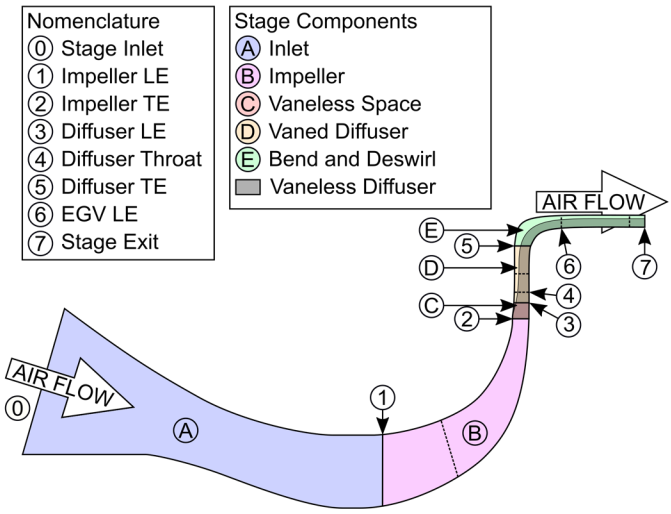


FIGURE 4: HECC VANED AND VANELESS DIFFUSER NOMENCLATURE

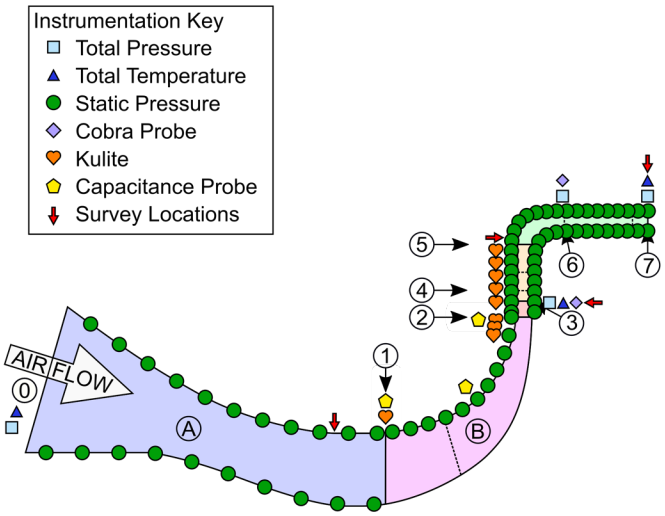


FIGURE 5: HECC VANED DIFFUSER STAGE INSTRUMENTATION OUTLINE.

instrumentation, notable observations during the experiments, stage geometry, and the structure of the directories in the data set, as well as necessary details for utilization of the data, such as column header descriptions.

4. GENERAL NOTES

4.1 Convention

The coordinate system for the HECC data set is cylindrical with the impeller axis of rotation serving as the axis of rotation. Per the convention in the report published by Medic et al. [1], the axial coordinate is 0 at the impeller leading edge and increasing in the forward-looking-aft (FLA) direction. Unless otherwise specified, the length units are in inches. The azimuthal direction is oriented such that top dead center (TDC) is equivalent to 0° and positive counterclockwise is oriented from the FLA direction, which corresponds to the direction of the impeller rotation.

4.2 Nomenclature and Instrumentation Overview

To give context for the details of the instrumentation presented in later sections, meridional illustrations of the HECC vaned and vaneless diffuser stages are provided for documentation of the stage nomenclature (Figure 4) and instrumentation (Figure 5). The station nomenclature is consistent between the vaned and vaneless diffuser stages for ease of comparison between the configurations. The total temperature and total pressure measurements at Station 0, considered the stage inlet, are located upstream of the illustrated meridional cross sections; specifically, they are located in the inlet plenum called out in Figure 1 and Figure 2. Station 0, Station 4 (the diffuser vane leading edge) and Station 7 (the stage exit), are the primary locations for characterization of the stage and component performance with total pressure and total temperature rakes.

All instrumentation channel names begin with letters that give some context for the measurement, Table 1. The few one-off exceptions to the convention documented in

TABLE 1. INSTRUMENTATION CHANNEL CONVENTION: MEASUREMENT TYPE

Letter Series	Description
P	Pressure
PEV	Total pressure measured via an instrumented EGV in the axial deswirl; at the EGV leading edge
PMV	Total pressure measured via an instrumented diffuser vane; at the vane leading edge
PO	Orifice plate pressure
T	Total temperature
TC	Tip clearance
TM	Metal temperature
TO	Orifice plate temperature

TABLE 2. INSTRUMENTATION CHANNEL CONVENTION: MEASUREMENT GROUP

Group	Description
00	Inlet plenum stagnation measurements
11	Inlet shroud statics, group 1
12	Inlet shroud statics, group 2
13	Inlet hub statics, group 1
14	Inlet hub statics, group 2
21	Impeller shroud statics, group 1
22	Impeller shroud statics, group 2
24	Impeller trailing edge statics
26	Impeller backplate statics, group 1
27	Impeller backplate statics, group 2
31	Vaneless space shroud statics
32	Diffuser shroud statics
35	Vaneless space hub statics
36	Diffuser hub statics
61	Bend shroud statics
62	Bend hub statics
63	EGV shroud statics
64	EGV hub statics
70, 71	Stage exit stagnation measurements

Table 1 are provided in the relevant *metadata* files outlined in the following sections. The first two numeric characters following the letters generally indicate the measurement group, Table 2. Stagnation conditions are measured exactly at the callout for the stage nomenclature in Figure 4, i.e., P70XX is a total pressure measured at the stage exit.

It should be noted that there are sometimes slight differences between the defined measurement location of static pressure taps in the *metadata* files and the hub or shroud boundary defined by the computational coordinates in the archive. These deviations are known, and most noticeable when comparing the hub and shroud boundaries of the inlet and vaneless diffuser with the static pressure taps within those components. In the inlet, the cause of the difference has not been fully determined, but the issue is under investigation. In the vaneless diffuser, the static tap locations were acquired from fabrication drawings at the cold conditions, so the difference there is attributed to the cold-to-hot deformation, given that the vaneless diffuser boundaries are defined at hot conditions.

4.3 Bleed Flows and Metal Temperatures

The locations of the external casing metal temperatures (TMEXxx) are shown on the test article in the photographs in Figure 6. TMEX01 is the furthest upstream on the inlet (visible in Figure 6(a)), and TMEX02 through TMEX05 are distributed from just downstream of the impeller knee through the trailing edge of the diffuser (Figure 6(b)). TMEX06 (not shown in the photographs in Figure 6) is located on the exterior of the exhaust collector (Figure 7). In addition to showing the external metal temperatures (TMEXxx series) in pink, Figure 7 presents the diffuser surface metal temperatures (TM42XX series) and EGV surface metal temperature (TMEV1) in yellow and the backplate metal temperatures (TM2XXX series) in turquoise. Finally, the backface bleed flow path is visible from the impeller trailing edge, between the impeller backface and the backplate assembly, and across the labyrinth seal.

The mass flow rate through the backface bleed flow path is not measured directly. However, the differential pressure across the labyrinth seal at the impeller backface is recorded at all times. This value is documented in the parameter CLSDP. A positive value of CLSDP indicates flow out from the main flow path exiting across the labyrinth seal. Typically, CLSDP is maintained at a value of approximately 0.5 psid. In some cases, the value is set to 0.0 psid with the goal of reducing the assumptions in numerical simulations, which do not account for bleed flow.

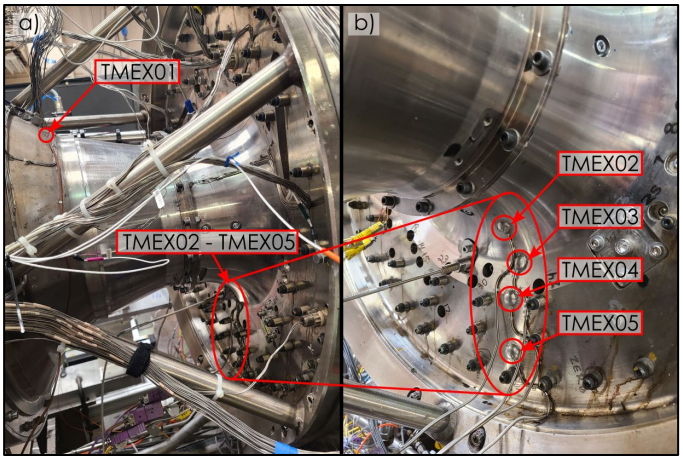


FIGURE 7: HECC CASING EXTERNAL METAL TEMPERATURES.

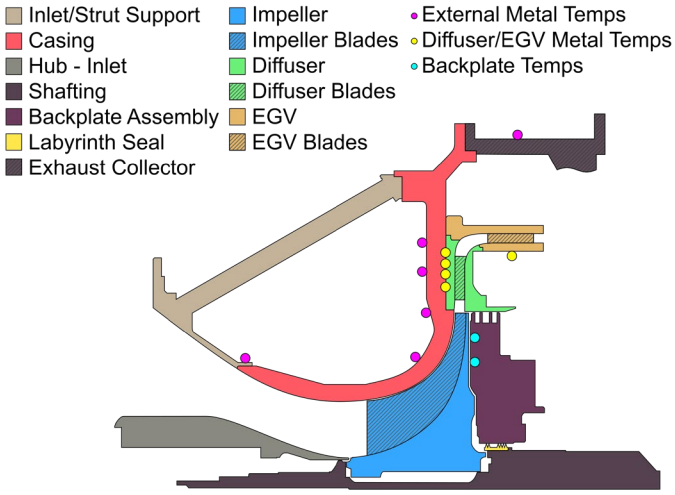


FIGURE 8: METAL TEMPERATURES AND BACKFACE BLEED FLOW PATH IN THE HECC TEST ARTICLE.

4.4 Gas Properties

Real gas properties with accounting for humidity were used for calculation of any given performance metrics or corrected parameters. The calculations were conducted in REFPROP [8] according to the procedures given by Berdanier et al. [9] and Lou et al. [10].

In Fall 2024, ambiguities were identified in the measurement of water content via relative humidity for the HECC Vaned Diffuser Data Set (Section 6), which resulted in undesirable variations in corrected speed for data along a constant speed characteristic. To improve the quality of the published data set, weather data from the Cleveland Hopkins Airport, which is located adjacent to the NASA Glenn Research Center, was used to determine

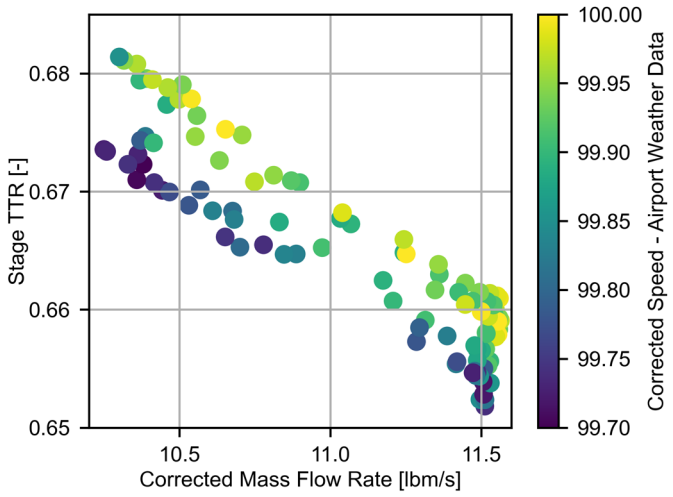


FIGURE 6: EFFECT OF HUMIDITY ON STAGE TOTAL TEMPERATURE RISE AT 100 PERCENT CORRECTED SPEED.

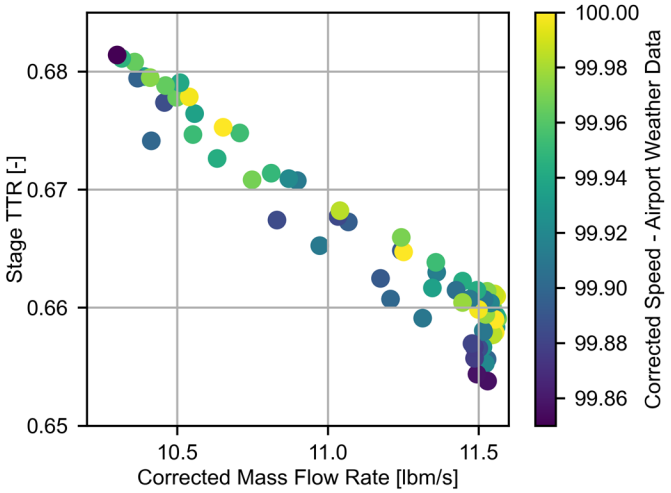


FIGURE 9: HECC VANED DIFFUSER DATA SET AT 100 PERCENT CORRECTED SPEED WITH TIGHT TOLERANCE IN CORRECTED SPEED PRESCRIBED.

the mass fraction of water in the atmospheric air at the time of each recording in the data set. The effect of the water content in the air is presented in Figure 8. Although the intent was for all the data in Figure 11 to be acquired at 100 percent corrected speed, inconsistencies in correcting for humidity resulted in an undershoot of up to 0.3 percent and undesirable variation in the data collected along the speedline.

As of the October 2024B Revision, the corrected conditions for the data set have been updated to use the airport data, which primarily impacts the corrected speed calculations due to changes in the molar composition of

the air as a function of humidity. Corrected mass flow rate and efficiency calculations are also impacted, though to a lesser degree because they are also dependent upon the corrected speed of the machine. Using a tighter tolerance band of acceptable variation in corrected speed, the scatter in the data set has been reduced (Figure 9).

Ultimately, for users of the HECC Data Archive, the performance metrics of the stage are unchanged, but the variation in performance along a speedline has been reduced.

4.5 Tip Clearance

The impeller tip clearance is measured with capacitance probes at three locations in the HECC: inlet, knee, and exducer (Figure 10). At each meridional location, there is a single tip clearance probe, and the runtime value of tip clearance is set based on the single tip clearance value at the exducer. However, rub probes are used to calibrate the tip clearance probes as well as to evaluate the eccentricity of the impeller clearance. Typical values of the eccentricity of the rub probes at 100 percent corrected speed are given in Figure 10. The measured value of tip clearance at the exit (`TCEX1`) is set to compensate for the eccentricity. As such, refer to the channel `TCEXSET` for the "set" value of the tip clearance. The difference between `TCEX1` and `TCEXSET` for each data set will be documented throughout the relevant sections of this document. Further details on the tip clearance measurements are provided in Reference [7].

4.6 Design-Intent and As-Manufactured Geometry Differences

Detailed investigations presented by Robles Vega et al. [6] have identified significant differences between the intended design blade sections of the HECC impeller (provided in the HECC Vaned Diffuser Report [1]) and the actual, manufactured impeller geometry. A summary of the findings is as follows:

1) The physical, manufactured HECC impeller is different from the intended design geometry (referred to hereafter as the "Design-Intent Geometry") presented by Medic et al. [1] with significant differences in the splitter leading edge metal angle, impeller exit radius, and blade thickness. These differences are too great to be attributed to errors in the hot-to-cold transformation process.
2) The physical, manufactured HECC impeller (referred to hereafter as the "As-Manufactured Geometry) is accurately represented at cold, unloaded conditions by the solid models contained in the HECC Data Archive.
3) Numerical simulations show that the Design-Intent Geometry achieves improved performance relative to the As-Manufactured Geometry.

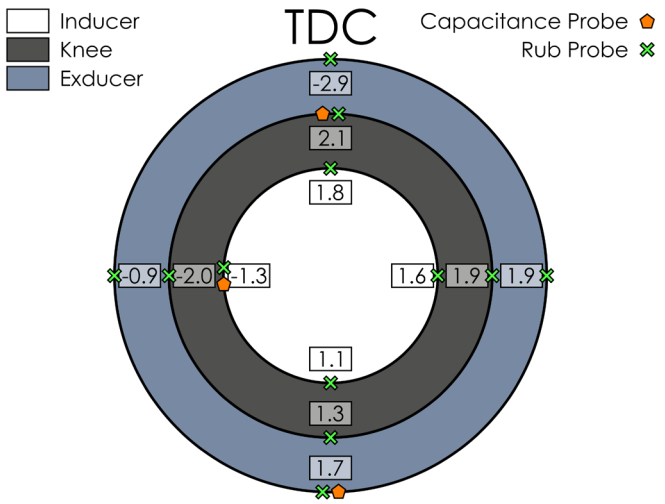


FIGURE 10: FLA ILLUSTRATION OF TIP CLEARANCE INSTRUMENTATION AND TIP GAP ECCENTRICITY IN THOUSANDTHS OF AN INCH AS MEASURED BY RUB PROBES AT 100 PERCENT CORRECTED SPEED DURING THE HECC TRANSITION DUCT TEST CAMPAIGN.

Users of the Data Archive are strongly encouraged to, at minimum, read the Abstract and Conclusions sections by Robles Vega et al.

Guidelines for use of the HECC geometry are as follows:

1) The Design-Intent Geometry was used for the pretest performance predictions of HECC. It is the intended shape of the impeller at hot, loaded conditions.
2) The As-Manufactured Geometry is the actual, fabricated impeller geometry that has been experimentally tested. All experimental results are for the As-Manufactured Geometry.
3) The impeller solid model contained in the data archive is an accurate representation of the As-Manufactured Impeller at cold, unloaded conditions.

The discrepancies between the Design-Intent and As-Manufactured geometries appear to be isolated to the impeller. If further discrepancies are observed, please

TABLE 3. KEY HECC IMPELLER DESIGN-INTENT AND AS-MANUFACTURED GEOMETRIC PARAMETERS. VALUES MARKED BY ASTERISKS WERE LIMITED TO APPROXIMATION BY CAD TOOLS DUE TO THE COMPLEX CURVATURE OF THE IMPELLER GEOMETRY.

Parameter	Design-Intent	As-Manufactured	Unit
LE Hub Radius	1.595	1.592	in.
LE Tip Radius	4.240	4.238	in.
Splitter LE Hub Radius	2.658	2.630	in.
Splitter LE Tip Radius	4.737	4.737	in.
Hub TE Radius	8.494	8.480	in.
Shroud TE Radius	8.499	8.482	in.
Average Exit Radius	8.497	8.481	in.
Exit Blade Height	0.609	0.600	in.
Tip Clearance	0.012	0.012	in.
Blade Number (Main/Splitter)	15/15	15/15	—
Backsweep	30	30	deg
LE Lean Angle	-2	-2*	deg
TE Lean Angle	-29	-29*	deg

contact Trey Harrison at herbert.harrison@nasa.gov to notify NASA of the issue.

5. KEY PARAMETERS

5.1 Geometric Parameters

Major parameters of interest, such as blade numbers, leading edge tip radius, and blade heights are compiled in this section. The Design-Intent and As-Manufactured impeller parameters are given in Table 3. The Design-Intent parameters are taken directly from Medic et al. [1]; the As-Manufactured parameters were extracted from a three-dimensional scan of the HECC impeller. The major parameters for the vaned diffuser and EGV are given in Table 4 and Table 5, respectively. Finally, the vaneless diffuser parameters at the impeller exit rating station and stage exit are given in Table 6.

TABLE 4. KEY HECC VANED DIFFUSER GEOMETRIC PARAMETERS.

Parameter	Value	Unit
MB Setting Angle from Radial	80	deg
Inlet Radius	9.108	in.
Exit Radius	11.199	in.
Span at LE	0.559	in.
Blade Number (Main/Splitter)	20/20	—

TABLE 5. KEY HECC EGV GEOMETRIC PARAMETERS.

Parameter	Value	Unit
Inlet Hub Radius	11.8315	in.
Inlet Shroud Radius	12.297	in.
Exit Hub Radius	11.868	in.
Exit Shroud Radius	12.305	in.
Axial Chord	2.438	in.
Blade Number	60	—

TABLE 6. KEY HECC VANELESS DIFFUSER GEOMETRIC PARAMETERS.

Parameter	Value	Unit
Impeller Exit Rating Plane – Radius	9.108	in.
Impeller Exit Rating Plane – Span	0.481	in.
Stage Exit – Hub Radius	11.869	in.
Stage Exit – Shroud Radius	12.182	in.
Stage Exit – Span	0.313	in.

5.2 Design Point Aerodynamic Conditions

At standard day conditions, defined as 59 °F, 14.7 psia, and 0 percent relative humidity, the design speed of the HECC impeller is 21,789 rpm with an exit tip clearance of 0.012 in. Medic et al. [1] specify the design point for the vaned diffuser stage in terms of exit corrected mass flow rate as 3 lbm/s. At design speed, an exit corrected mass flow rate of 3 lbm/s corresponds to an inlet corrected

mass flow rate of 10.85 lbm/s, at which the stage delivers a total pressure ratio of 4.68 at an isentropic efficiency of 82.1 percent. For the vaneless diffuser stage, the design inlet mass flow rate is considered to be the peak efficiency point on the 100 percent characteristic. Peak efficiency for the vaneless diffuser stage was measured at an inlet mass flow rate of 11 lbm/s and total pressure ratio 4.55. The measured isentropic efficiency of the vaneless stage was 83.8 percent with the baseline metal inlet and 82.6 percent with the baseline plastic inlet, which implies a notable effect from heat transfer. See Reference [3] for information regarding the effect of the inlet on the stage performance.

Given that the vaneless diffuser stage design point is at peak efficiency for the vaneless diffuser as well as the impeller [3,7], it is the opinion of the author that an inlet corrected mass flow rate of 11 lbm/s is a reasonable operating condition for analysis of the HECC impeller, vaned diffuser, and vaneless diffuser components.

6. HECC VANED DIFFUSER DATA SET

The *HECC Vaned Configuration* directory contains the *Coordinates*, *Data*, and *Models* subdirectories, which contain blade section information, experimental data, and solid models, respectively, associated with the HECC vaned diffuser configuration. The convention for the data set is given in Section 4.1 Convention.

6.1 HECC Vaned Stage Component Coordinates

6.1.1 HECC Design-Intent Stage Coordinates (Hot, No Fillets)

The *HECC Design-Intent Stage Coordinates (Hot, No Fillets)* directory contains the Design-Intent hub, shroud, impeller main blade, impeller splitter blade, diffuser main vane, diffuser splitter vane, and exit guide vane (EGV) coordinates of the HECC vaned diffuser configuration. The Design-Intent Impeller differs from the As-Manufactured Impeller as discussed in Section 4.6 Design-Intent and As-Manufactured Geometry Differences. The differences between the As-Manufactured and Design-Intent geometries are isolated to the impeller, and there is good agreement between Design-Intent and As-Manufactured geometries for the diffuser vanes and the EGVs.

The component defined by the files is explicitly included in each file name. The blade sections as given in the original report published by Medic et al. [1] and reproduced here do not include the fillets that are present in the actual hardware on the impeller blades, diffuser vanes, and EGVs. The fillets on the hardware can be viewed in the solid model discussed in Section 6.4 Models. The coordinates account for the expected deformation of the components associated with the temperatures and loadings of the stage at design conditions.

Except for the flow path hub and shroud files, the coordinates are space delimited with three columns. A single line header defines the columns: `X`, `R*THETA`, and `R`, which correspond to the axial, radius*angle, and radial directions. Specifically, the `R*THETA` column is the radial coordinate multiplied by the azimuthal coordinate in radians. The blade sections are specified in single complete loops for each file, with the lowest increment in the filename corresponding to the hub location and increasing increment moving towards the shroud.

The flow path hub and shroud files contain only two columns and are tab delimited. The column headers `x` and `r` are the axial and radial coordinates, respectively, given in inches and radians.

6.1.2 HECC As-Manufactured Impeller Coordinates (Cold, With Fillets)

The *HECC As-Manufactured Impeller Coordinates (Cold, With Fillets)* directory contains the Cold As-Manufactured impeller blade section data directly extracted from the impeller solid model contained in the Data Archive. These blade sections are at cold, unloaded conditions and are derived from the solid model that was used for fabrication of the stage. As such, the Cold, As-Manufactured coordinates do not account for the deformation of the impeller that occurs at high speeds and loadings. The impeller blade sections are given in the same convention as specified in Section 6.1.1 HECC Design-Intent Stage Coordinates (Hot, No Fillets).

6.1.3 HECC As-Manufactured Impeller Coordinates (Hot, With Fillets)

The *HECC As-Manufactured Impeller Coordinates (Hot, With Fillets)* directory contains the As-Manufactured impeller blade section data at the hot, loaded conditions at 100 percent corrected speed. These coordinates were obtained by conducting a fluid-structure interaction study with coupled finite element and computational fluid dynamics (CFD) analysis to determine the deformation of the As-Manufactured impeller geometry from cold, unloaded conditions to hot, loaded conditions. The deformation is most notable near the trailing edge, where the total displacement is

approximately 0.040 in. The As-Manufactured Hot impeller coordinates are recommended for use in numerical simulations of the HECC at 100 percent corrected speed. The Hot As-Manufactured impeller blade sections are given in the same convention as specified in Section 6.1.1 HECC Design-Intent Stage Coordinates (Hot, No Fillets).

The cold-to-hot transformation of the As-Manufactured impeller geometry was conducted by Qingyuan Zhuang at Siemens, and the resulting Hot As-Manufactured impeller coordinates have been generously provided to NASA to host on the Data Archive. The procedures and analyses of the cold-to-hot transformation will be published in a forthcoming paper documenting the effect of the deformation from cold, unloaded conditions to hot, loaded conditions on the stage performance.

6.2 Data

The *Data* directory contains multiple files. Critical to interpretation of the actual data is the *HECCvaned_metaData.csv*, which contains metadata associated with the instrumentation and calculated parameters of the experiment in comma-delimited format. Excluding the single line header, each row is either instrumentation from the experiment or a calculated quantity. The single line column header has entries with data in each column as follows:

- `CN`: channel name corresponding to value in data files
- `DESCRIPTION`: description of the channel
- `UNITS`: units of the channel
  - `DEG F`: degrees Fahrenheit
  - `DEG R`: degrees Rankine
  - `MIL`: thousandths of an inch
  - `PSIA`: absolute pressure in lbf/in^2
  - `PSID`: differential pressure in lbf/in^2
  - `RPM`: rotations per minute
- `CAT`: category of the channel
  - `AERO` for aerodynamic instrumentation
  - `CALC` for calculated parameter
  - `FLAG` for noting data purpose or instrumentation configuration
- `SCAT`: subcategory for instrumentation type
  - `DP`: differential pressure
  - `MISC`: miscellaneous
  - `PS`: static pressure
  - `PT`: total pressure
  - `RH`: relative humidity
  - `RPM`: rotational speed
  - `TC`: tip clearance
  - `TM`: metal temperature
  - `TT`: total temperature
- `R`: radial coordinate of instrumentation in inches
- `THETA`: azimuthal coordinate of instrumentation in degrees (see Section 6 for convention)
- `Z`: axial coordinate of instrumentation in inches (see Section 6 for convention)
- `SPAN`: spanwise location of instrumentation in percent, increasing from hub=0 percent to shroud=100 percent
- `DPITCH`: diffuser pitch location of instrumentation in percent increasing from suction side to pressure side of the diffuser vane (in the direction of impeller rotation)
- `EGVPITCH`: EGV pitch location of instrumentation in percent increasing from suction side to pressure side of the diffuser vane (in the direction of impeller rotation)

Blank fields in the *HECCvaned_metaData.csv* spreadsheet are generally not relevant to the associated parameter.

Of particular note in the meta data are the `RDG`, `DFLAG`, and `CFLAG` parameters. Each data entry has a unique identifier referred to as a reading number that is stored in the `RDG` column of the data spreadsheet. The `DFLAG` parameter is an integer flag identifier to signal the type of reading recorded. `CFLAG` is also an integer flag identifier used to note the configuration of the compressor at the time of the respective reading. For example, this may indicate the presence or absence of survey probes or modular vanes. The integer values for the `DFLAG` and `CFLAG` used in the publicly available data sets hosted in the archive at the time of writing and their associated meanings are given in Table 7 and Table 8.

6.2.1 Tip Clearance

Due to the eccentricity of the impeller relative to the shroud, the tip clearance at the impeller trailing edge varies around the circumference of the exducer, and the tip clearance value recorded in channel `TCEX1` is 0.003 in. less than the maximum clearance around the circumference. During all vaned diffuser testing discussed in Section 6, `TCEX1` was set such that the maximum clearance value around the circumference was equivalent

TABLE 3. HECC VANED DFLAG (DATA FLAG) INTERPRETATION.

DFLAG Value	Meaning
1	Steady-state speedline recording
8	Stage inlet survey

TABLE 4. HECC VANED CLFAG (CONFIGRUATION FLAG) INTERPRETATION.

CLFAG Value	Meaning
1	Modular vane total pressure rakes installed at position 1 (MODVN1) and position 2 (MODVN2)
6	Cobra-style three-hole probe installed in the inlet duct upstream of the impeller

to the intended value for clearance. The intended value for clearance is saved in the channel TCEXSET. TCEXSET is obtained by adding the difference between TCEX1 and the maximum clearance around the circumference (in this section, 0.003 in.).

As an example, if the intended value for a reading is 0.012 in., the impeller would be translated axially until the value for TCEX1 was equal to 0.009 in. Because the difference between TCEX1 and the maximum clearance value is 0.003 in., the maximum clearance would be 0.012 in., and the value of 0.012 in. would be recorded in the channel TCEXSET. In practice, the recorded values in these channels are usually within 0.0005 in. of the target value. Further details on the tip clearance variation in the vaned diffuser testing documented herein are given by Braunscheidel et al. [4].

6.2.2 Experimental Data

The experimental data is stored in separate files based on tip clearance setting. As noted in Section 6.2.1, the tip clearance setting is the maximum tip clearance around the circumference and is stored in channel TCEXSET.

In the present revision, detailed experimental data at the design tip clearance is provided in the *HECCvanedData_12MilExitClearance.csv* file, and the compressor map in Figure 11 gives an overview of the data contained therein. The channels and descriptions in the data are included in the previously described *HECCvaned_metaData.csv* file.

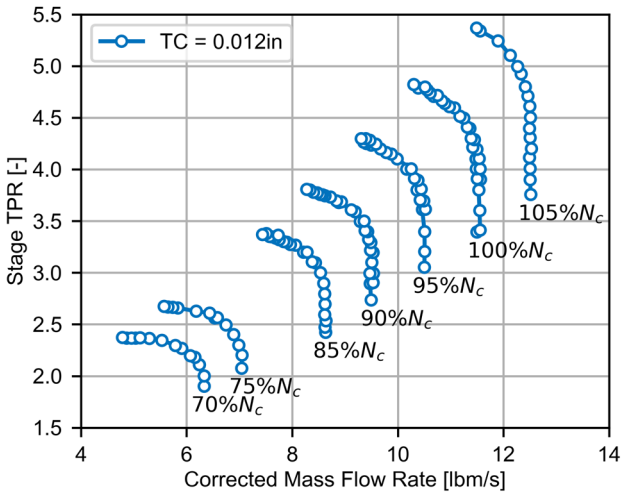


FIGURE 11: HECC VANED DIFFUSER CONFIGURATION COMPRESSOR MAP AT 0.012" IMPELLER TRAILING EDGE CLEARANCE.

6.3 Data – Stage Inlet Surveys

Stage inlet surveys were conducted with a cobra-style three-hole probe upstream of the impeller to quantify the flow angle and total pressure provided to the impeller. The surveys were acquired at numerous operating conditions throughout the compressor map (Figure 12). The target operating conditions for many of the surveys was an exit corrected mass flow rate of 3 lbm/s, which can be observed in Figure 13, where the stage total pressure ratio is given as a function of the stage exit corrected mass flow rate. The surveys are obtained by traversing the probe across the span of the inlet duct at an axial coordinate of -1.00 inches. The compressor is maintained at the same operating condition throughout the survey. An example of processed survey data is given in Figure 14. The *Data* directory contains the *Stage Inlet Surveys* directory, which in turn contains the stage inlet survey files.

The filenames within the directory indicate the operating condition at which the surveys were collected and are in the following format:

inletFlowAngleSurvey_XXXNc_XXexitTC_X.XlabSeal_XX.XXmdot_X.XXTPR.csv

Each series of characters between the underscores gives a specification of the operating conditions, and each X character is a numeric character giving values for each of the performance parameters within the name. The type of survey is initially given, followed by the percent corrected speed in three digits (a leading 0 is present at speeds less than 100 percent), tip clearance set point in thousandths

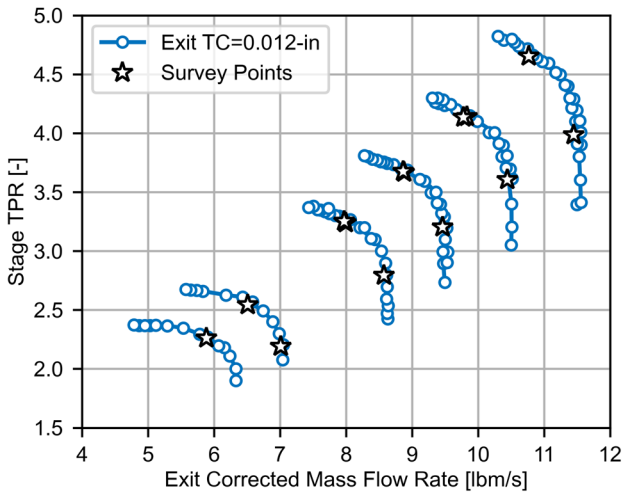


FIGURE 12: HECC VANED DIFFUSER CONFIGURATION MAP WITH INLET SURVEY LOCATIONS INDICATED.

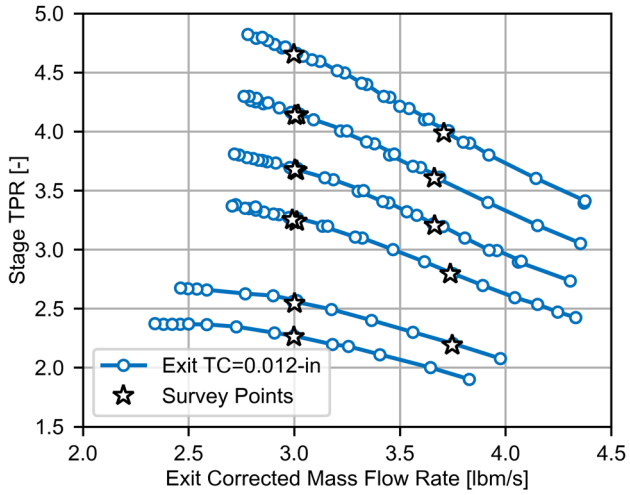


FIGURE 13: HECC VANED DIFFUSER CONFIGURATION MAP IN TERMS OF EXIT CORRECTED MASS FLOW RATE WITH INLET SURVEY LOCATIONS INDICATED.

of an inch, labyrinth seal differential pressure (see Sections 4.3 and 7.3.2 Baseline Experimental Data Sets) corrected mass flow rate, and stage total pressure. As an example, the file *inletFlowAngleSurvey_099Nc_13exitTC_0.5labSeal_11.45mdot_5.02TPR.csv* is stage inlet flow angle survey recorded at 99 percent corrected speed, an exducer exit clearance of 0.013 in., a labyrinth seal differential pressure of 0.5 psid, corrected mass flow rate of 11.45 lbm/s, and a total pressure ratio of 5.02.

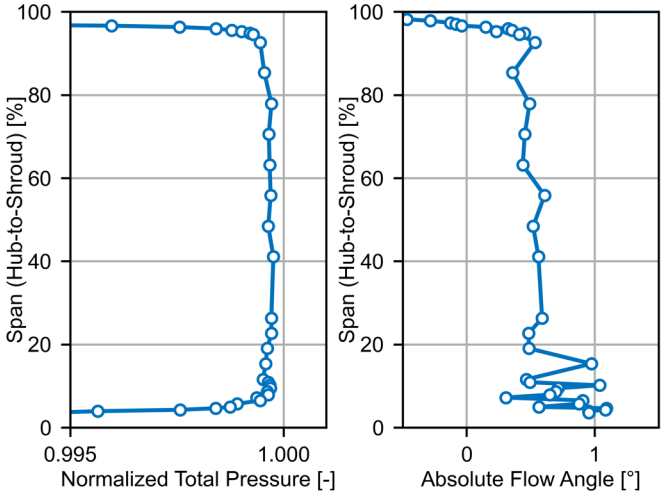


FIGURE 14: HECC INLET SURVEY DATA AT 100 PERCENT CORRECTED SPEED, 10.77 LBM/S INLET MASS FLOW RATE.

TABLE 5. HECC VANED INLET SURVEY CHANNELS OVERVIEW (SEE METADATA FILE FOR DETAILS).

Channel	Summary
PROBESPAN	Probe location in percent span from hub (0 percent) to shroud (100 percent)
INLETPT	Calculated value of total pressure from the inlet survey probe
INLETALPHA	Calculated value of inlet absolute flow angle from the inlet survey probe

The data within each of the survey files is structured similarly to the previously discussed steady-state data, and only a few small changes are incorporated to accommodate the inclusion of survey data. The relevant probe survey channels are outlined in Table 9 with additional details included in the *HECCvaned_metaData.csv* file, the structure of which is detailed in Section 6.2 Data.

Each individual survey recording is identified by a unique reading number given in the survey data files. Generally, the location of the probe in the PROBESPAN, INLETPT, or INLETALPHA channels will change with each reading to build the survey of data across the span of the inlet duct. In addition to the unique reading numbers in the RDG column, a MEAN row is provided. The MEAN row is the average of the steady performance parameters

recorded for each unique reading within the survey to give the average operating conditions over the entire survey acquisition process.

General notes on the survey data files:

- Any values for span data greater than 100 percent (`PROBESPAN` channel) indicate the probe is recessed within the shroud.
- The `MEAN` values are not applicable for the survey channels themselves, as the probe location changes throughout the survey.
- The processed flow angle data is obtained through the method given by Treaster and Yocum [11].

6.4 Models

A solid model of the HECC vaned diffuser stage is available in the *Models* directory in the form of a step file, *HECCvaned_AsManufacturedStage_cleanGeometry.stp*. The model is a simplified version of the original detailed model of the stage. Extraneous features, such as instrumentation and components beyond the bounds of the stage, are removed. The fillets on the blades of the impeller, vaned diffuser, and EGV are all included. Unfortunately, due to the complexity of the impeller model, the impeller often contains defects after exporting from various modelling packages. To help mitigate this issue, a separate impeller model is included in the *HECC_AsManufacturedImpeller_cleanGeometry.stp* file. Both models provide the As-Manufactured geometry of the compressor stage and components at cold, unloaded conditions.

7. HECC VANELESS DIFFUSER DATA SET

The *HECC Vaneless Configuration* directory contains the *Coordinates*, *Data – Baseline Metal Inlet*, *Data – Transition Duct Inlet*, and *Models* directories. Although the inlet configuration and diffuser configuration are different from the vaned diffuser configuration, the impeller is unchanged. The convention for the data set is consistent with the other data sets in the archive and is documented in Section 4.1 Convention. A review of the vaneless diffuser performance is provided by Harrison, McNichols, and Blaha [3].

7.1 Coordinates

The *Vaneless Hub and Shroud - Baseline Metal Inlet (Hot)* directory (located in the *Coordinates* directory contains the hot hub and shroud profiles of the vaneless diffuser configuration with the baseline metal (BLM) inlet. The impeller is unchanged throughout the configurations, so the impeller blade sections are not repeated in the vaneless diffuser directory. However, they can be retrieved from the vaned configuration blade sections as documented in Section 6.1 HECC Vaned Stage Component Coordinates.

7.2 Data – Baseline Metal Inlet

The *Data – Baseline Metal Inlet* directory is structured in a fashion similar to that of the vaned diffuser data directories. Critical to interpretation of the actual data is the *HECCvaneless_baselineMetalInlet_metaData.csv*, which contains metadata associated with the instrumentation and calculated parameters of the experiment in comma-delimited format. Excluding the single line header, each row is either instrumentation from the experiment or a calculated quantity. The single line column header has entries with data in each column as follows:

- `CN`: channel name corresponding to value in data files
- `DESCRIPTION`: description of the channel
- `UNITS`: units of the channel
  - `DEG F`: degrees Fahrenheit
  - `DEG R`: degrees Rankine
  - `MIL`: thousandths of an inch
  - `PSIA`: absolute pressure in lbf/in^2
  - `PSID`: differential pressure in lbf/in^2
  - `RPM`: rotations per minute
- `CAT`: category of the channel
  - `AERO` for aerodynamic instrumentation
  - `CALC` for calculated parameter
  - `FLAG` for noting data purpose or instrumentation configuration
- `SCAT`: subcategory for instrumentation type
  - `DP`: differential pressure
  - `MISC`: miscellaneous
  - `PS`: static pressure
  - `PT`: total pressure
  - `RH`: relative humidity
  - `RPM`: rotational speed
  - `TC`: tip clearance
  - `TM`: metal temperature
  - `TT`: total temperature
- `R`: radial coordinate of instrumentation in inches
- `THETA`: azimuthal coordinate of instrumentation in degrees (see Section 6 for convention)

- Z: axial coordinate of instrumentation in inches (see Section 6 for convention)
- SPAN: spanwise location of instrumentation in percent increasing from hub=0 percent to shroud=100 percent
- SETTINGANGLE: angle of the instrumentation relative to axial direction (only relevant to stage exit total pressure rakes)

Blank fields throughout the *HECCvaneless_baselineMetalInlet_metaData.csv* spreadsheet are generally not relevant to the associated parameter.

Of particular note in the metadata are the RDG, DFLAG, and CFLAG parameters. Each data entry has a unique identifier referred to as a reading number that is stored in the RDG column of the data spreadsheet. The DFLAG parameter is an integer flag identifier to signal the type of reading recorded. CFLAG is also an integer flag identifier used to note the configuration of the compressor at the time of the respective reading. For example, this may indicate the presence or absence of survey probes or modular instrumentation. For the vaneless diffuser, various types of modular instrumentation may be installed from the shroud via access ports. The options for modular instrumentation are actuated probes, which can be traversed across the span, total pressure rakes, total temperature rakes, and a series of static pressure taps.

The integer values for the DFLAG and CFLAG used in the publicly available data sets hosted in the archive at the time of writing and their associated meanings are given in Table 10 and Table 11. For the different configurations indicated by CFLAG, the channels in the data files are unchanged. This results in cases where the channels exist but do not contain valid data. For clarity, for the channels that are not connected during testing but still exist in the file, the entries in the column are empty.

7.2.1 Tip Clearance

Due to the eccentricity of the impeller relative to the shroud, the tip clearance at the impeller trailing edge varies around the circumference of the exducer in the same manner as in the vaned diffuser testing. The details of the tip clearance measurement are given in Section 4.5 Tip Clearance and Section 6.2.1 Tip Clearance. The parameter TCEXSET contains the intended value of tip clearance at the impeller exit, which in the vaneless diffuser BLM data set is the maximum tip clearance measured around the circumference according to the rub probes.

TABLE 6. HECC VANELESS BASELINE METAL INLET **DFLAG** (DATA FLAG) INTERPRETATION.

DFLAG Value	Meaning
1	Steady-state speedline recording

TABLE 7. HECC VANELESS BASELINE METAL INLET **CLFAG** (CONFIGURATION FLAG) INTERPRETATION.

CLFAG Value	Meaning
8	Modular vane static pressure tap series installed (PMV3 and PMV4)

7.2.2 Experimental Data

The experimental data is stored in separate files based on tip clearance setting. As noted in Section 7.2.1 Tip Clearance, the tip clearance setting is the maximum tip clearance around the circumference and is stored in channel TCEXSET.

In the present revision, detailed experimental data at the design tip clearance is provided in the *HECCvanelessData_baselineMetalInlet_12MilExitClearance.csv* file, and the compressor map in Figure 15 gives an overview of the data contained therein. The channels and descriptions in the data are included in the *HECCvaneless_baselineMetalInlet_metaData.csv* file previously described.

Finally, as noted by Harrison [3], the total temperature rakes at the impeller exit failed before reaching high speeds in the vaneless diffuser test campaign. As such, there are no total temperature measurements available at the impeller trailing edge. However, the stage total temperature rakes were functional through the entire test campaign. Additionally, the impeller exit total temperature rakes were redesigned for the transition duct test, and that data was successfully acquired and recorded. The impeller exit total temperature data is available in the transition duct data files.

7.3 Data – Transition Duct Inlet

The *Data – Transition Duct Inlet* directory is structured in a fashion similar to that of the vaneless diffuser data directories. At the present revision, a single metadata file, *HECCvaneless_tDuct_metaData.csv*, provides the information for the instrumentation and calculated parameters in the experimental data. Excluding the single line header, each row is either instrumentation

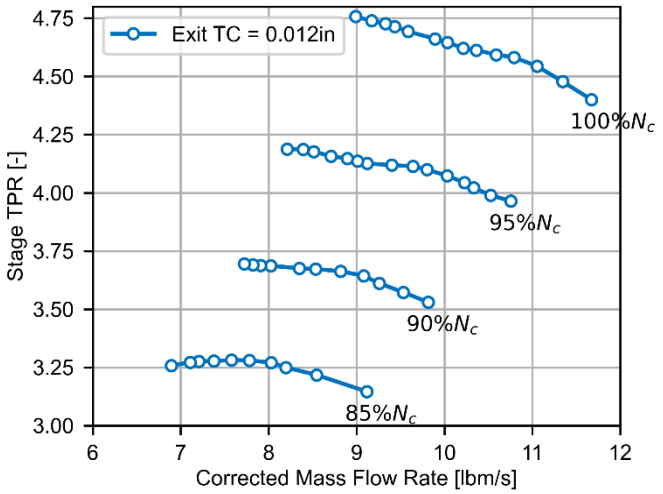


FIGURE 15: HECC VANELESS DIFFUSER BLM CONFIGURATION COMPRESSOR MAP AT 0.012-INCH IMPELLER TRAILING EDGE CLEARANCE.

from the experiment or a calculated quantity. Each column has headers and entries as follows:

- `CN`: channel name corresponding to value in data files
- `DESCRIPTION`: description of the channel
- `UNITS`: units of the channel
  - `DEG F`: degrees Fahrenheit
  - `DEG R`: degrees Rankine
  - `MIL`: thousandths of an inch
  - `PSIA`: absolute pressure in lbf/in^2
  - `PSID`: differential pressure in lbf/in^2
  - `RPM`: rotations per minute
- `CAT`: category of the channel
  - `AERO` for aerodynamic instrumentation
  - `CALC` for calculated parameter
  - `FLAG` for noting data purpose or instrumentation configuration
- `SCAT`: subcategory for instrumentation type
  - `DP`: differential pressure
  - `MISC`: miscellaneous
  - `PS`: static pressure
  - `PT`: total pressure
  - `RH`: relative humidity
  - `RPM`: rotational speed
  - `TC`: tip clearance
  - `TM`: metal temperature
  - `TT`: total temperature
- `R`: radial coordinate of instrumentation in inches
- `THETA`: azimuthal coordinate of instrumentation in degrees (see Section 6 for convention)
- `Z`: axial coordinate of instrumentation in inches (see Section 6 for convention)
- `SPAN`: spanwise location of instrumentation in percent increasing from hub=0 percent to shroud=100 percent
- `SETTINGANGLE`: angle of the instrumentation relative to axial direction (only relevant to stage exit total pressure rakes)

Blank fields throughout the *HECCvaneless_tDuct_metaData.csv* spreadsheet are generally not relevant to the associated parameter.

Key metadata for the transition duct are the `RDG`, `DFLAG`, and `CFLAG` parameters, and although the information for each of the flags is quite similar to that in Section 7.2 Data – Baseline Metal Inlet, the content is repeated here for convenience and clarity. Additionally, the transition duct data contains an additional flag, `TDFLAG`, which indicates the configuration of the additively manufactured transition duct inlet. All transition duct inlet configurations are manufactured from Stratasys's acrylonitrile styrene acrylate (ASA). The values and meanings of `TDFLAG` are given in Table 12, and a detailed review of the transition duct configurations can be found in Harrison et al. [3]. For the modified hub inlet (`TDFLAG` =1), some static pressure taps along the hub of the inlet flow path are removed. In the case where those channels are invalid, the entries in the column are empty.

Each data entry has a unique identifier referred to as a reading number that is stored in the `RDG` column of the data spreadsheet. The `DFLAG` parameter is an integer flag identifier to signal the type of reading recorded. `CFLAG` is also an integer flag identifier used to note the configuration of the compressor at the time of the respective reading. For example, this may indicate the presence or absence of survey probes or modular instrumentation. For the vaneless diffuser, various types of modular instrumentation may be installed from the shroud via access ports. The options for modular instrumentation are actuated probes that can be traversed across the span, total pressure rakes, total temperature rakes, and a series of static pressure taps.

The integer values for the `DFLAG` and `CFLAG` used in the publicly available data sets hosted in the archive at the time of writing and their associated meanings are given in Table 13 and Table 14. For the different configurations indicated by `CFLAG`, the channels in the data files are

TABLE 8. HECC VANELESS TRANSITION DUCT INLET `TDFLAG` (DATA FLAG) INTERPRETATION.

TDFLAG Value	Meaning
0	Baseline inlet, additively manufactured ASA
1	Modified-hub inlet, additively manufactured ASA

TABLE 9. HECC VANELESS TRANSITION DUCT `DFLAG` (DATA FLAG) INTERPRETATION.

DFLAG Value	Meaning
1	Steady-state speedline reading
5	Impeller exit survey reading

TABLE 10. HECC VANELESS TRANSITION DUCT `CLFAG` (CONFIGRUATION FLAG) INTERPRETATION.

CLFAG Value	Meaning
0	Modular vane total pressure rakes are installed at impeller exit rating station (PMV1 and PMV2 series instrumentation)
1	Three-hole cobra-style probes are installed at the impeller exit rating station (MV1C and MV2C series instrumentation)
2	Total temperature probes are installed at the impeller exit rating station (MV1TT and MV2TT)

unchanged. This results in cases where the channels exist but do not contain valid data. For clarity, when the channels were not connected during testing but still exist in the file, the entries in the column are empty.

7.3.1 Tip Clearance

The eccentricity of the impeller relative to the shroud discussed in Section 6.2.1 Tip Clearance was improved in the transition duct testing. As such, the parameters `TCEX1` and `TCEXSET` are identical. Further details on the eccentricity of the tip clearance are given by Harrison et al. [7].

7.3.2 Baseline Experimental Data Sets

The experimental data is stored in separate files based on tip clearance and inlet configuration. In the present

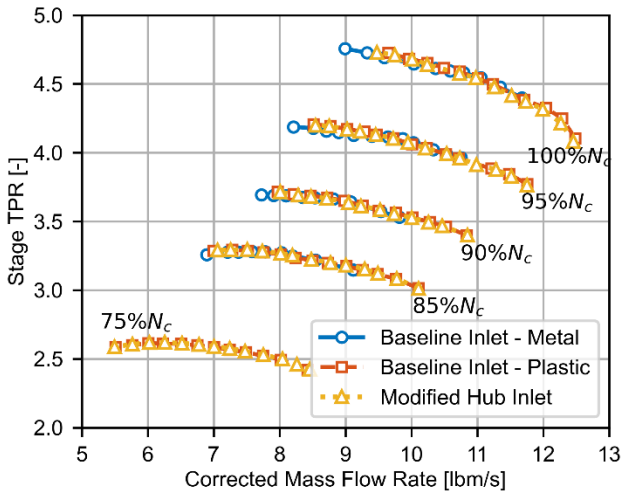


FIGURE 16: HECC VANELESS DIFFUSER COMPRESSOR MAP AT 0.012 IN. IMPELLER TRAILING EDGE CLEARANCE FOR EACH INLET CONFIGURATION.

revision, detailed experimental data at the design tip clearance for the baseline plastic inlet and modified hub inlet are provided in the *HECCtductData_TD00_12MilExitClearance.csv* and *HECCtductData_TD01_12MilExitClearance.csv* files, respectively. The compressor map in Figure 16 shows the overall performance of the vaneless diffuser configuration with the baseline metal inlet (BLM), baseline plastic (BLP), and modified hub (MH) inlet configurations. The difference in low (stalling) and high (choking) mass flow rates recorded for the BLM and BLP/MH testing is due only to testing decisions and is not derived from the change in inlet configuration itself. The low mass flow rate was limited by the throttle during BLM testing. The throttle was then adjusted to accommodate larger mass flow rates during the BLP/MH testing, which allowed the impeller to approach inducer choke at high speeds. Additionally, surge testing was conducted during the BLM test. To preserve the integrity of the plastic components, surge was avoided in the BLP/MH testing. The channels and descriptions in the data are included in the previously described *HECCtduct_metaData.csv* file.

Also included in the experimental data is the *HECCtductData_TD00_12MilExitClearance_CLSDP0.csv* file. This set of experimental data is recorded with the BLP inlet and vaneless diffuser as well, but the compressor labyrinth seal differential pressure (`CLSDP`) has been set to 0 psid. The labyrinth seal allows for bleed flow to pass

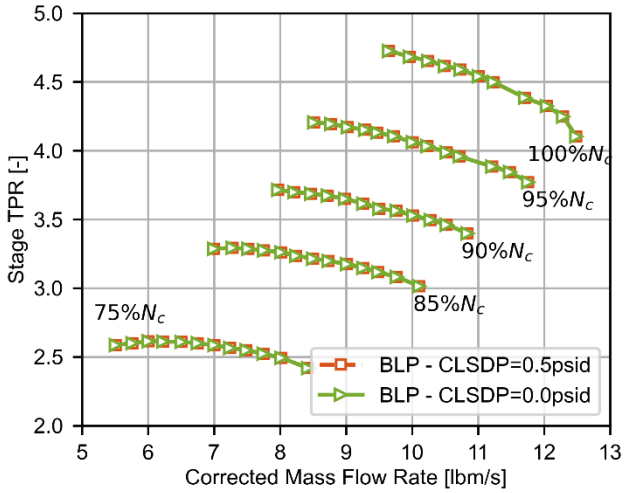


FIGURE 17: HECC VANELESS DIFFUSER BLP CONFIGURATION COMPRESSOR MAP AT 0.012 IN. IMPELLER TRAILING EDGE CLEARANCE FOR NOMINAL AND 0 PSID LAB SEAL DIFFERENTIAL PRESSURE.

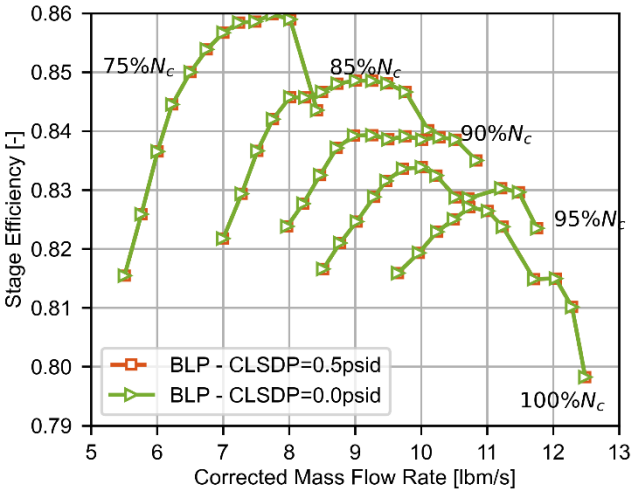


FIGURE 18: HECC VANELESS DIFFUSER BLP CONFIGURATION EFFICIENCY MAP AT 0.012 IN. IMPELLER TRAILING EDGE CLEARANCE FOR NOMINAL AND 0 PSID LAB SEAL DIFFERENTIAL PRESSURE.

behind the backface of the impeller and exit through the secondary air flow system (see Section 4.3 Bleed Flows and Metal Temperatures). In all previously discussed data sets, the labyrinth seal was set to 0.5 psid to provide air flow for cooling the backface region of the compressor stage: high pressure air at the impeller trailing edge passes across the backface and exits through the labyrinth seal at

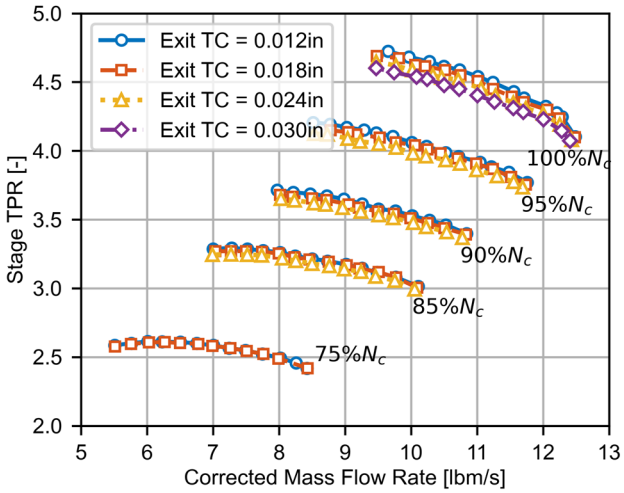


FIGURE 19: HECC VANELESS DIFFUSER BLP CONFIGURATION COMPRESSOR MAP FOR ALL ACQUIRED CLEARANCE CONDITIONS.

positive values for `CLSDP`. The *HECCtductData_TD00_12MilExitClearance_CLSDP0.csv* data set was recorded with the value for the labyrinth seal set to 0 psid such that there should be no flow out of the stage through the backface leak path. It is provided to give simplified operating conditions in which there is minimal secondary flow into or out of the stage. From 0.0 psid to 0.5 psid, the labyrinth seal bleed has a negligible impact on the stage compressor map in terms of stage total pressure ratio (Figure 17) and efficiency (Figure 18).

7.4 Data – Tip Clearance

The tip clearance data set is also contained in the *Data – Transition Duct Inlet* directory because the associated data were acquired with the BLP additively manufactured inlet configuration. Metadata associated with the tip clearance data set are the exact same as that given in the *HECCvaneless_tDuct_metaData.csv* file. As such, users of the archive are directed to Section 7.3 Data – Transition Duct Inlet for details on the configuration metadata.

Both steady-state speedline and impeller exit survey data were collected at tip clearance values of 0.012 in., 0.018 in., 0.024 in., and 0.030 in. The steady-state data are contained in the *HECCtductData_TD00_allClearance.csv* file. A compressor map of the tip clearance data set is given in Figure 19 for reference. As previously stated, the experimental data were acquired from near-choke to near-stall conditions. Instability was avoided to preserve the integrity of the plastic inlet components.

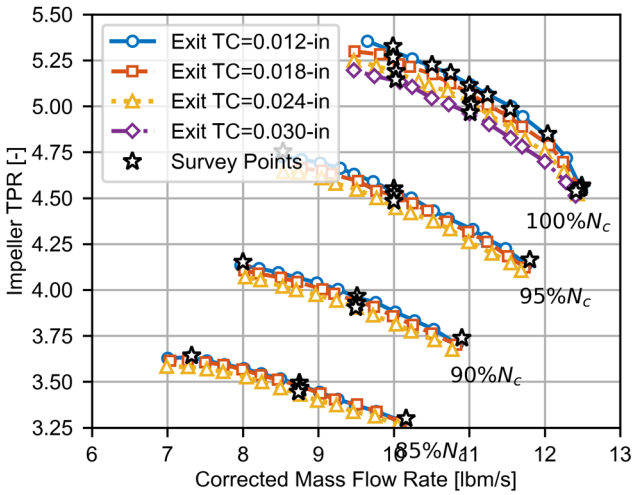


FIGURE 20: HECC IMPELLER MAP WITH OPERATING CONDITIONS AT WHICH IMPELLER EXIT FLOW ANGLE SURVEYS WERE COLLECTED INDICATED BY STARS.

Impeller exit surveys of flow angle and total temperature were also acquired at each tip clearance condition. The details for interpretation of the impeller exit surveys are provided in Section 7.5 Data – Impeller Exit Surveys.

7.5 Data – Impeller Exit Surveys

Impeller exit surveys of flow angle, total pressure, and total temperature were acquired at numerous operating conditions throughout the operating range of the compressor (Figure 20). The surveys were obtained by traversing probes across the span of the vaneless diffuser at the impeller exit rating plane, and the compressor was maintained at the same operating condition throughout the survey. Examples of processed survey data can be found in Reference [7].

The *Data – Transition Duct Inlet* directory contains the *Impeller Exit Surveys* directory, which in turn contains the *Flow Angle* and *Total Temperature* directories. As their names indicate, *Flow Angle* and *Total Temperature* directories contain impeller exit surveys of flow angle and temperature data across the span at the impeller exit rating station.

The filenames within each of the subdirectories indicate the operating condition at which the surveys were collected and are in the following format:

- *impellerExitFlowAngleSurvey_TDX_XXXNc_XXexitTC_X.XlabSeal_X.XXmdot_X.XXTPR.csv*
- *impellerExitTtSurvey_TDX_XXXNc_XXexitTC_X.XlabSeal_X.XXmdot_X.XXTPR.csv*

Each series of characters between the underscores gives a specification of the operating conditions, and each X character is a numeric character giving values for each of the performance parameters within the name. The type of survey is initially given, followed by the transition duct configuration (see Table 12), percent corrected speed in three digits (a leading 0 is present at speeds less than 100 percent), tip clearance set point in thousandths of an inch, labyrinth seal differential pressure (see Sections 4.3 and 7.3.2 Baseline Experimental Data Sets), corrected mass flow rate, and stage total pressure. For example, the file *impellerExitFlowAngleSurvey_TD0_085Nc_18exitTC_0.5labSeal_8.76mdot_3.47TPR.csv* is an impeller exit flow angle survey recorded with the baseline plastic transition duct configuration at 85 percent corrected speed, an exducer exit clearance of 0.018 in., a labyrinth seal differential pressure of 0.5 psid, corrected mass flow rate of 8.76 lbm/s, and a total pressure ratio of 3.47.

The data within each of the survey files is structured similarly to the previously discussed steady-state data, and only a few small changes are incorporated to accommodate the inclusion of survey data. The relevant probe survey channels are outlined in Table 15 with additional details included in the *HECCvaneless_tDuct_metaData.csv* file, the structure of which is detailed in Section 7.3 Data – Transition Duct Inlet.

Each individual survey recording is identified by a unique reading number given in the survey data files. Generally, the location of the probe in the `PACXX`, `MVXPANGLE`, or `MVXSPAN` channels will change with each reading to build the survey of data across the span at the impeller exit rating station. In addition to the unique reading numbers in the `RDG` column, a `MEAN` row is provided. The `MEAN` row is the average of the steady performance parameters recorded for each unique reading within the survey to give the average operating conditions over the entire survey acquisition process.

General notes on the survey data files:

- Any values for span data less than 0 percent (`PACXX` channels) or greater than 100 percent (`MVXSPAN` channels) indicate the probe is recessed within the shroud.
- The `MEAN` values are not applicable for the survey channels themselves, as the probe location changes throughout the survey.

TABLE 11. HECC VANELESS TRANSITION DUCT SURVEY CHANNELS OVERVIEW (SEE METADATA FILE FOR DETAILS).

Channel	Summary
PAC01, PAC03	Raw survey probe actuator angular position, measured in degrees relative to radial direction
PAC02, PAC04	Raw survey probe actuator linear position measured in percent span with shroud and hub corresponding to 0 percent and 100 percent span, respectively
MV1CC, MV1CL, MV1CR, MV2CC, MV2CL, MV2CR	Raw pressure data measured with each cobra-style three-hole probe measured at the modular vane 1 and 2 locations
MV1TT, MV1TT	Raw total temperature data measured with each cobra-style three-hole probe measured at the modular vane 1 and 2 locations
MV1PANGLE, MV1PT, MV2PANGLE, MV2PT, MV1ALPHA, MV2ALPHA, MV1SPAN, MV2SPAN,	Processed three-hole probe and/or total temperature probe data

- The processed flow angle data is obtained through the method given by Treaster and Yocum [11].

7.6 Models

The solid model of the HECC vaneless diffuser is provided in the *HECCvaneless_cleanGeometry.stp* file. As with other solid models in the Data Archive, the geometry is provided at cold, unloaded conditions.

8. REFERENCES

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